

Meridian — Richer timeline components (groups, roles, brackets, eras, typed events, media)

Context

Meridian renders family timelines as swimlanes. Today a person carries three kinds of sub-items — `events` (point diamonds), `periods` (filled role bars, e.g. "Mayor of Ploiești"), and `groups` (dashed background boxes, e.g. "Săhăteni estate") — plus project-level `annotations` (the  accolade *braces*). These come only from sample data / GEDCOM: there is **no create/edit/delete UI**, no color, no notes, no link to a source, and no stable IDs. The only interaction is a hover tooltip (plus a delete `×` on braces). "Research mode" is a single boolean that turns drag-to-create-a-brace on.

The user wants to author a richer timeline (inspiration: time.graphics):

- **Group & label multiple events** into thematic bands (Communist regime, a law in effect, an estate administered) — this is the existing `groups` layer, enriched.
- **Role/occupation labels kept separate** from those bands (e.g. "Mayor of Ploiești") — the existing `periods` layer, enriched.
- **Accolade brackets** over a span, not tied to a group or events — the existing `annotations` layer, enriched.
- Every one of these gets **title, color, start/end, and an optional reference to a source**.
- **View mode:** clicking any item opens a **detail modal** (even if it only has a date).
- **Research mode:** clicking opens an **editable modal** with delete; new items are created by **drag-selecting a span** → a "New item" type-picker modal.
- Plus three additional components: **full-board era bands**, **typed event markers** (icons/colors by `type` + legend), and **pinned media/photos** on a lane.

Intended outcome: a research-grade, editable timeline where every span/point is a first-class, colored, sourced, clickable object that persists via the existing IndexedDB autosave.

Design

1. Data model (add IDs + fields; new factories)

Give every sub-item a stable `id` and the new optional fields. Add factories to `src/models/project.js`, mirroring the existing `createSource` pattern (`crypto.randomUUID()` + defaults + `...rest`):

- `createEvent({ year, label, type='note', note='', color=null, sourceId=null })`
- `createPeriod({ start, end, label, note='', color=null, sourceId=null })` — role/occupation bar
- `createGroup({ start, end, label, note='', color=null, sourceId=null })` — event-group band
- `createAnnotation({ laneKey, start, end, title='', note='', color=null, sourceId=null })` — bracket
- `createEra({ start, end, label, color=null, note='', sourceId=null })` — **new** board-level band
- `createMediaPin({ year, sourceId, label='', color=null })` — **new** per-person media pin

Add `eras: []` to the project shape in `createProject` (parallel to `annotations`). Add `media: []` to the per-person shape (populated only via the new UI/normalization).

Migration: add `normalizeProject(project)` to `src/models/project.js` that (a) ensures `project.eras` exists, and (b) walks every person filling missing `ids` on events/periods/groups/media and ensuring those arrays exist. Call it in `useProject` right after `loadProject /seed` and in `openProject`, so the already-seeded "Hariton Family" project and any GEDCOM imports upgrade transparently. Also add IDs (and a couple of demo colors + a `SAMPLE_ERAS` export) to `src/sampleData.js` so a fresh seed showcases the features.

2. Type registries & palette (`src/theme.js`)

- `EVENT_TYPES`: map of `type → { label, icon, color }` for marriage / office / property / note / death-note / media (icons are plain unicode glyphs to match the existing 📄 / ★ aesthetic).
- `ITEM_COLORS`: a small swatch array (reuse `C.rel` + `C.focus` / `C.slate`) for the color picker.
- Reuse existing `hexA`, `btnStyle`, `C`.

3. New `useProject` operations (`src/hooks/useProject.js`)

- `setEras(fn)` — project-level, identical shape to `setAnnotations`.
- `updatePerson(id, fn)` — convenience over `setPeople` so App can add/edit/delete a person's events/periods/groups/media by id without repeating the map/replace boilerplate.
- Wire `normalizeProject` into the load/seed/open paths.

4. Interaction model (`App.jsx`)

- **Clicks vs pan:** interactive items get `onPointerDown={e => e.stopPropagation()}` (so the board's pan / annotation-drag in `onPointerDown` at `src/App.jsx:382` never starts) plus an `onClick` that opens the modal. A shared helper builds these props + `cursor:pointer`.
- **View mode** (`research === false`): click → `ItemModal` in **read-only** detail form.
- **Research mode** (`research === true`): click → `ItemModal` in **editable** form (with Delete).

- **Create:** repurpose the existing drag flow (`onPointerDown / Move / Up` , `draft` band at `src/App.jsx:387-416`). On pointer-up over a **person lane** with a real span, open the new type-picker create modal instead of today's note-only modal. A plain **click** (no drag) in research mode on a person lane opens the modal defaulted to a point **Event** at that year. Context lanes (`ctx:*` , sourced from the constant `CONTEXT_LANES` , not persisted) only offer **Bracket** and **Era**; person lanes offer Event / Role / Group / Bracket / Era / Media.

5. ItemModal (new `src/components/ItemModal.jsx`)

One component built on the existing `Modal` (`src/components/Modal.jsx`) handling create / edit / view via an `editable` flag:

- **Type picker** (create only): Event · Role · Group · Bracket · Era · Media (filtered by lane).
- **Fields:** title/label; start+end (or single year for Event/Media) prefilled from the drag, editable; color swatches (`ITEM_COLORS`); note textarea; **Source** `<select>` populated from `project.sources` (optional). Media type additionally requires picking an image/doc source.
- **View mode:** same layout rendered as static text; shows the linked source with an **Open** action — for `url` sources a link, for file sources load via `storage.loadBlob(blobKey)` → object URL (revoked on close), showing an inline `<img>` preview for images.
- Buttons: Save / Cancel, and Delete when editing. Routing on save: Event/Role/Group/Media → `updatePerson(laneId, ...)` ; Bracket → `setAnnotations` ; Era → `setEras` .

6. Rendering additions (App.jsx)

- **Era bands:** render inside the existing full-board underlay layer (`src/App.jsx:569-575` , `top:0 bottom:0 left:HEADER_W right:0`) as translucent colored vertical bands with a top label chip; band non-interactive, chip `pointerEvents:auto` + click → detail/edit.
- **Group bands** (`src/App.jsx:187-193`): keep the dashed box; apply the group's `color` ; make the label chip clickable (box stays background).
- **Role bars** (`PeriodBar` , `src/App.jsx:134-143`): apply `period.color` ; add click.
- **Event diamonds** (`EventDot` , `src/App.jsx:146-159`): color by `EVENT_TYPES[type].color` (fallback to lane color), show the type icon in the caption/tooltip; add click.
- **Braces** (`Brace` , `src/App.jsx:161-173`): apply `color` ; show `title` ; click opens detail; keep `×` delete but gate it to research mode.
- **Media pins:** new small `MediaPin` component — a thumbnail (loads `sourceId` 's blob via `storage.loadBlob` in an effect → object URL) or  marker on the lane; click → detail preview.
- **Legend:** a "Legend" toolbar button opening a small `Modal` listing `EVENT_TYPES` (icon + swatch + label).
- Reuse `packRows` (`src/App.jsx:30`) for role-bar stacking as today.

Critical files

- `src/models/project.js` — new factories, `eras` default, `normalizeProject` .
- `src/sampleData.js` — IDs, demo colors, `SAMPLE_ERAS` .
- `src/theme.js` — `EVENT_TYPES` , `ITEM_COLORS` .
- `src/hooks/useProject.js` — `setEras` , `updatePerson` , `normalize` on load/open/seed.
- `src/components/ItemModal.jsx` — **new** unified create/edit/detail modal.
- `src/App.jsx` — click helper, create-flow type picker, era/group/role/event/brace/media rendering + clicks, legend button, view-vs-research gating.

Verification

1. `npm run build` and `npm test` (extend `src/tests/models.test.js` for the new factories + `normalizeProject` ; the existing suite covers `models/storage/useProject`).
2. `npm run dev` → <http://localhost:5173/meridian/> and, on the Hariton sample:
 - **View mode:** click a role bar, group band, event diamond, and bracket → detail modal shows title/date(s)/note and, where set, the linked source (open link / image preview).
 - **Research mode:** drag across the focus lane → type-picker modal → create a Role, a Group, and a Bracket with color + source; a plain click creates a point Event; add a full-board **Era** band (e.g. Communist regime 1947–1989) and confirm it spans all lanes; **pin a media** source and see its thumbnail; edit and delete items.
 - Confirm typed-event colors/icons render and the **Legend** button lists them.
3. Reload the page to confirm everything persisted through IndexedDB autosave (dirty dot clears), and that the pre-existing seeded project upgraded via `normalizeProject` without errors.